

Marco Barbierato

Experience

- 2025 – Now **Data Scientist - L2**, *Prima Assicurazioni*, Milan
April
Data Scientist - L3 (Senior) from April 2026
A/B testing. Overseeing Junior Colleagues.
R&D of features for frequency and severity of claims. Developing the feature platform data integration and quality assurance tests, serving 300k requests/day.
- I helped my team migrate from a proprietary GLM software to an open-source pipeline based on MLflow and XGBoost.
 - Rewrote the feature platform unit tests with PyTest utils to improve scalability.
- Working in a remote-first team of ~15 data scientists and ML engineers.
- 2024 – 2024 **Internship - National Institute of Astrophysics**, Observatory of Padova
March July
I developed Computer Vision Deep Learning models with PyTorch for the super-resolution of images taken by a hyper-panoramic camera, used for the exploration of lunar lava caves.
- I demonstrated models perform better than classic interpolation methods on human-like quality metrics.
 - I showed with an xAI technique why Transformer models performed better than CNNs despite standard metrics preferring the latter.
- I contributed to the open-source development of NeoSR, a Python package for super resolution.

Education

- 2022 – 2024 **Master's Degree in Data Science**, *Università degli Studi di Padova*, 110/110 with Honors
Department of Mathematics. Study Plan 'Mathematics of Machine Learning'.
Thesis: Super-resolution and deblurring of hyper-panoramic images using neural networks.
- **University Tutor** for the Masters course 'Fundamentals of Information Systems' (Fall '24). I held lectures for a class of 20 students on data structures, algorithms, and Python/SQL programming.
- 2019 – 2022 **Bachelor's Degree in Mathematics**, *Università degli Studi di Padova*, 104/110
Thesis: Bada wasins within the escape dynamics of Hamiltonian systems.

Publications

- Submitted for Review Barbierato M., et al. "Super-Resolution of Hyper-Hemispherical Images suffering from Spatially Variant Blur".
- 2025 Simioni, E., et al. "Pin-holes Based Description of the PANCAM, a Hyper-hemispherical Lens." *Optics, Photonics and Lasers* (2025).

Miscellaneous

- I am proficient in Python and numpy, pandas, scikit-learn, and PyTorch for scientific machine learning. Working knowledge of PyTest, SQL, PySpark, MLFlow.
- 2026 I made some small contributions (link) to Salsa-CI, the continous integration pipeline of Debian's Gitlab instance.
- 2024 PINN-Pad - Physics Informed Neural Networks in Padova - Workshop attendee
- 2018 Cambridge First Certificate in English B2 - Performance at Level C1 (182)